Final Exam

Question 1: Water Balance Analysis

You are analyzing a medium-sized catchment ($A = 450\ \mathrm{km}^2$) over a one-year period. The catchment experiences a Mediterranean climate with distinct wet and dry seasons. Monthly water balance data has been collected, though some components are incomplete.

Given data:

Month	Precipitation P (mm)	Runoff Q (mm)	Storage Change ΔS (mm)	Evaporation E (mm)
Jan	125	85	+15	?
Feb	142	98	+22	?
Mar	178	134	+18	?
Apr	92	68	-8	?
May	48	32	-12	?
Jun	28	15	-6	?
Jul	15	8	-4	?
Aug	22	12	-5	?
Sep	38	24	-3	?
Oct	68	45	-2	?
Nov	95	62	+5	?
Dec	108	72	+12	?

Table 1

Tasks:

a) Using the water balance equation $P = Q + E + \Delta S$, calculate the monthly evaporation for each month. Show your calculations for at least three months (one from each season: wet, transition, dry).

b) Identify which months represent the wet season, transition period, and dry season based on the data patterns. For each season, identify which term in the water balance equation is dominant and explain why.

c) In May, the storage change measurement is uncertain. If the actual evaporation in May is known to be 28 mm, use the water balance equation to calculate the corrected storage change. Discuss what this correction implies about the catchment's behavior during this period.

d) A neighboring catchment of similar size ($A = 480\ \mathrm{km}^2$) but in an arid climate shows the following annual totals: $P = 280\ \mathrm{mm}$, $Q = 45\ \mathrm{mm}$, $E = 210\ \mathrm{mm}$, and $\Delta S = 25\ \mathrm{mm}$. Compare the annual water balances of the two catchments and explain how climate differences manifest in the relative magnitudes of the balance components.

Question 2: Orographic Precipitation and Measurement Errors

A hydrological study is being conducted in a mountainous region with strong elevation gradients. Four rain gauges have been installed at different elevations, but measurement errors need to be quantified and corrected before calculating areal precipitation.

Gauge locations and characteristics:

Gauge	Elevation (m asl)	Shield Type	Wind Speed (m/s)	Measured Rainfall (mm)
A	450	Standard	4.2	245
B	820	Standard	6.8	328
C	1250	Shielded	5.5	412
D	2100	Shielded	7.2	487

Table 2

Additional information:

- Wind correction factor for standard gauges: $f_w = 1 - 0.04 \times \text{wind speed}$ (for wind speeds > 3 m/s)

- Wind correction factor for shielded gauges: $f_w = 1 - 0.02 \times \text{wind speed}$ (for wind speeds > 3 m/s)
- Elevation correction: Precipitation increases by 12% per 100 m elevation gain above 500 m asl
- Base elevation for correction: 500 m asl

Tasks:

a) Using the isohyetal method, construct isohyets (precipitation contours) at intervals of 50 mm (200 mm, 250 mm, 300 mm, 350 mm, 400 mm, 450 mm). You may sketch or describe the pattern. Then calculate the areal average precipitation using the isohyetal method formula: $R = \frac{1}{A}\sum_{i=1}^{n} R_i \times a_i$, where R_i is the average precipitation between adjacent isohyets, a_i is the area between those isohyets, and A is the total catchment area. For simplicity, assume the catchment area is $85\ \text{km}^2$ and the areas between isohyets are: 200-250 mm band: 12 km², 250-300 mm band: 18 km², 300-350 mm band: 22 km², 350-400 mm band: 20 km², 400-450 mm band: 8 km², 450+ mm band: 5 km².

b) For each gauge, calculate the wind-induced measurement error. Determine which gauge experiences the largest absolute error (in mm) and which experiences the largest relative error (as percentage).

c) Apply wind corrections to the measured rainfall values. Then, if you were to use a simple arithmetic mean of the corrected gauge readings, what would be the catchment average? Compare this to your isohyetal result and discuss which method is more appropriate for this orographic setting.

d) Gauge B is located in a valley that experiences a rain shadow effect from a nearby peak. How might this affect the gauge's representativeness, and what additional consideration should be given when interpreting its measurements in the context of orographic precipitation patterns?

Question 3: Evaporation Across Different Environments

Evaporation processes vary dramatically across different climatic and environmental settings. Understanding these variations is critical for water resource management.

Three contrasting environments:

Environment 1: Arid desert region

- Mean annual precipitation: 180 mm
- Mean annual potential evaporation: 2800 mm
- Deep water table (> 50 m)
- Sparse vegetation (desert shrubs)
- Sandy, well-drained soils

Environment 2: Humid temperate forest

- Mean annual precipitation: 1400 mm
- Mean annual potential evaporation: 850 mm
- Shallow water table (2-5 m)
- Dense forest canopy
- Clay-rich, moisture-retentive soils

Environment 3: Cold alpine tundra

- Mean annual precipitation: 450 mm (mostly snow)
- Mean annual potential evaporation: 320 mm
- Permafrost present
- Low-lying vegetation (mosses, lichens)
- Shallow, organic-rich soils

Tasks:

a) For each of the three environments, compare the likely relationship between actual evaporation (E_t) and potential evaporation (E_{pt}). Which environment will have the smallest ratio E_t/E_{pt} and which will have the largest? Provide quantitative estimates and explain your reasoning.

b) Explain how the three fundamental controls on evaporation (available energy, water availability, and atmospheric demand) operate in each environment. Identify which control is most limiting in each case and why.

c) Evaluate which evaporation estimation method (Thornthwaite temperature-based, Penman combination, or water balance approach) would be most appropriate for each environment. Justify your choice by considering data availability, accuracy requirements, and method assumptions. Discuss the limitations of each method in the contexts where it is less suitable.

d) At the catchment scale, direct measurement of evaporation is extremely challenging. Discuss three major obstacles to obtaining reliable catchment-average evaporation estimates, and explain how these obstacles vary between the three environments described above.

Question 4: Runoff Hydrograph Analysis

A stream gauging station has recorded a complex hydrograph following a series of rainfall events over a 10-day period. The hydrograph shows multiple peaks resulting from different storm events and varying runoff generation mechanisms.

Hydrograph data:

Time (days)	**Discharge Q (m^3/s)**	**Cumulative Rainfall (mm)**
0.0	2.5	0
0.5	3.2	8
1.0	15.8	32
1.5	28.4	45
2.0	22.1	45
2.5	12.3	52
3.0	8.7	52
3.5	18.2	72
4.0	31.5	85
4.5	26.8	85
5.0	15.4	85
5.5	19.3	95
6.0	33.1	108
6.5	28.9	108
7.0	11.2	108

7.5	6.8	108
8.0	4.5	108
8.5	3.9	108
9.0	3.2	108
9.5	2.8	108
10.0	2.6	108

Table 3

Catchment characteristics:

- Catchment area: $A = \mathbf{125\ km}^2$
- Initial baseflow: 2.5 m^3/s
- Soil type: Clay-loam with moderate infiltration capacity
- Land use: Mixed agriculture and forest
- Antecedent conditions at day 0: Moderately wet (soil moisture at 65% of field capacity)

Tasks:

a) Identify the three main storm events (by time period) and estimate the peak discharge for each. For each storm, calculate the time to peak from the start of rainfall. Separately estimate the direct runoff volume and delayed/baseflow contribution for the first storm (days 0-2.5). Assume baseflow decays exponentially: $Q_b(t) = Q_{b0}e^{-kt}$ where $k = \mathbf{0.15\ day}^{-1}$ and $Q_{b0} = \mathbf{2.5\ m}^3/\mathrm{s}$.

b) Calculate the runoff coefficient ($C = Q_{\text{runoff}}/P$) for each of the three storm events, where Q_{runoff} is the direct runoff volume converted to mm depth. Compare the runoff coefficients and explain why they might differ between storms.

c) Based on the hydrograph shapes, antecedent conditions, and catchment characteristics, identify which runoff generation mechanism (infiltration-excess overland flow, saturation overland flow, or subsurface contributions) is most likely dominant. Justify your answer by referencing specific features of the hydrograph (peak timing, recession behavior, baseflow contribution).

d) If the same rainfall pattern occurred during a drought period when antecedent soil moisture was only 25% of field capacity, predict how the hydrograph would differ. Specifically, discuss changes in: (i) peak discharge magnitude, (ii) time to peak, (iii) runoff coefficient, and (iv) baseflow contribution. Provide quantitative estimates where possible.

Question 5: Layered Aquifer Flow Analysis

A contaminant spill has occurred near a multi-layered aquifer system, and you need to analyze groundwater flow paths and estimate travel times for risk assessment.

Aquifer system characteristics:

The system consists of four horizontal layers (from top to bottom):

Layer	Thickness b (m)	Hydraulic Conductivity K (m/day)	Porosity n	Type
1	8	25	0.32	Sand
2	3	0.8	0.28	Silt
3	15	18	0.30	Sand
4	5	0.05	0.22	Clay

Table 4

Flow conditions:

- Total hydraulic head at upgradient boundary: $h_1 = 125$ m
- Total hydraulic head at downgradient boundary: $h_2 = 118$ m
- Distance between boundaries: $L = 1200$ m
- Contamination point: Located at the upgradient boundary, top of Layer 1
- Aquifer width (perpendicular to flow): $W = 500$ m

Tasks:

a) Calculate the equivalent horizontal hydraulic conductivity (K_h) and equivalent vertical hydraulic conductivity (K_v) for the entire layered system. Explain why $K_h \neq K_v$ and what this means for flow behavior.

b) Assuming horizontal flow dominates, calculate the horizontal hydraulic gradient and the Darcy velocity in each layer. If flow is primarily horizontal, in which layer would you expect the highest flow velocity?

c) Calculate the transmissivity for the entire system (sum of $T_i = K_i \times b_i$ for all layers). Then, assuming the system behaves as a single confined aquifer with this transmissivity, calculate the total volumetric flow rate (Q) using: $Q = T \times W \times i$, where i is the hydraulic gradient.

d) Estimate the travel time for a conservative (non-reactive) contaminant moving from the spill location to the downgradient boundary, assuming it follows horizontal flow paths. Use the linear velocity equation $v = \frac{K \times i}{n_e}$, where n_e is the effective porosity (assume $n_e = 0.9 \times n$ for sands and $n_e = 0.85 \times n$ for finer materials). Which layer will control the overall travel time, and why?

Question 6: Aquifer Characterization and Management

A groundwater consultant has collected data from a new well field development site. Your task is to analyze the data to characterize the aquifer system and evaluate its management potential.

Field measurement data:

Well A (near recharge area):

- Ground surface elevation: 450 m above sea level
- Well casing extends to 80 m depth
- Water level in well (depth below surface): 12 m
- Screened interval: 65-75 m depth
- Pressure measurement at screen midpoint (70 m depth): 520 kPa (absolute)

Well B (downgradient, near discharge area):

- Ground surface elevation: 442 m above sea level
- Well casing extends to 90 m depth
- Water level in well (depth below surface): 28 m
- Screened interval: 80-90 m depth

- Pressure measurement at screen midpoint (85 m depth): 380 kPa (absolute)

Regional information:

- Estimated annual recharge: $R = 150$ mm/yr over catchment area of 85 km^2
- Current pumping rate from 12 production wells: $Q_{\text{pump}} = 8.5 \times 10^6$ m^3/yr
- Estimated annual discharge to streams: $Q_{\text{discharge}} = 2.2 \times 10^6$ m^3/yr
- Aquifer material: Medium to fine sand
- Top confining layer begins at 25 m depth
- Bottom of aquifer unit at 100 m depth

Additional data:

- Water density: $\rho_w = 1000$ kg/m^3
- Gravitational acceleration: $g = 9.81$ m/s^2

Tasks:

a) For both wells, calculate the hydraulic head at the screened interval. Determine whether this aquifer system is confined or unconfined, and justify your conclusion using the head measurements, pressure data, and well construction information.

b) Explain the fundamental difference in storage mechanisms between confined and unconfined aquifers. Given that both storage mechanisms (water compressibility and aquifer matrix compressibility for confined; gravity drainage for unconfined) operate, why do we observe such different values of storativity ($S = 10^{-5}$ to 10^{-3} for confined vs. $S = 0.01$ to 0.30 for unconfined)? Use specific storage concepts in your explanation.

c) Based on the recharge and discharge data, evaluate whether the current pumping rate is sustainable. Calculate the annual water balance and determine if there is a storage deficit. If pumping continues at this rate, what might happen to the hydraulic heads and what are the long-term implications?

d) This aquifer type (unconsolidated sand) differs significantly from other major aquifer types such as karst limestone or fractured crystalline rock. Compare the management challenges for: (i) contamination risk and remediation, (ii) sustainable yield estimation, and (iii) monitoring network design. For each challenge category, explain why the approach differs between unconsolidated and the other aquifer types you discuss.

Question 7: Satellite Geodesy for Water Cycle Monitoring

Remote sensing via satellite missions has revolutionized our ability to monitor water cycle components at regional to global scales. Understanding what satellites actually measure versus what hydrological quantities they derive is essential for proper interpretation.

Three satellite missions for water monitoring:

Mission 1: GRACE (Gravity Recovery and Climate Experiment) and GRACE Follow-On

- Orbital altitude: ~500 km
- Measurement: Micrometer-level changes in distance between two satellites
- Spatial resolution: ~300 km
- Temporal resolution: Monthly
- Physical principle: Measures Earth's gravity field variations

Mission 2: SMAP (Soil Moisture Active Passive)

- Orbital altitude: ~685 km
- Measurement: Microwave brightness temperature at L-band (1.4 GHz)
- Spatial resolution: ~40 km (passive), ~3 km (active)
- Temporal resolution: 2-3 days global coverage
- Physical principle: Measures electromagnetic emission from land surface

Mission 3: SWOT (Surface Water and Ocean Topography)

- Orbital altitude: ~891 km
- Measurement: Water surface height using radar interferometry
- Spatial resolution: ~50-200 m (swath-dependent)
- Temporal resolution: Variable (near-polar orbit, repeat cycle)
- Physical principle: Measures radar phase differences between two antennas

Tasks:

a) For each of the three missions (GRACE, SMAP, SWOT), clearly identify: (i) what physical property is directly measured by the satellite sensors, and (ii) what hydrological quantity is derived from these measurements using retrieval algorithms or models. Explain why satellites cannot directly measure most hydrological variables.

b) Explain the physical principles that underlie each measurement approach. For GRACE, discuss how mass changes (water storage) affect gravity. For SMAP, explain how soil moisture affects dielectric properties and microwave emission. For SWOT, describe how radar interferometry measures water surface elevation.

c) For each mission, compare satellite-derived estimates with traditional ground-based measurement methods. Identify two advantages and two limitations of the satellite approach relative to ground measurements. Consider factors such as spatial coverage, temporal resolution, accuracy, cost, and data accessibility.

d) In practice, satellite data are often integrated with ground-based observations through data assimilation or hybrid monitoring strategies. Discuss when and why such integration is necessary. Provide a specific example of how you would combine GRACE total water storage data with in-situ groundwater well measurements to improve estimates of regional groundwater changes. What are the key challenges in this integration?